

Appendix II – Tables

Table 3.1 – Raw Variables Descriptive Statistics

Variable	N	Mean	Std	Min	25%	50%	75%	Max
M1 (JPY)	767	2.80e14	2.85e14	4.02e12	5.80e13	1.52e14	4.83e14	1.08e15
M2 (JPY)	746	3.66e14	3.04e14	3.56e12	4.82e13	3.10e14	6.48e14	9.63e14
M3 (JPY)	527	9.36e14	3.32e14	2.95e14	7.62e14	9.93e14	1.13e15	1.60e15
Total Credit – Priv. Non-Fin. (%GDP)	732	161.885	26.993	110.800	139.175	160.850	179.075	213.600
Total Credit – Gen. Gov. (%GDP)	336	172.047	39.522	88.300	142.200	183.250	206.800	231.500
Total Credit – HH & NPISHs (%GDP)	732	53.638	14.944	19.900	44.325	60.150	64.950	70.200
Total Treasury Reserves (– Gold)	837	4.15e5	5.11e5	5.98e2	1.33e4	8.18e4	9.83e5	1.37e6
10-Year Gov. Bond Yield (%)	446	1.811	1.823	-0.280	0.569	1.334	1.954	8.032
Call Money/Interbank Rate (%)	488	1.246	2.183	-0.071	0.001	0.091	0.766	8.278
Est. 1-Year Neutral Interest Rate (%)	372	0.354	0.580	-0.420	-0.261	0.244	0.864	1.380
Real GDP (bn chained 2015 JPY)	384	535407	36656	461492	503948	535754	568025	593800
Central Gov. Debt (%GDP)	385	129.955	57.276	38.176	80.478	128.600	192.113	216.135
Priv. Debt Securities (%GDP)	277	61.365	8.777	46.448	55.154	59.193	68.580	78.847
Pub. Debt Securities (%GDP)	277	140.771	37.064	63.163	111.205	157.153	176.610	217.026
BoJ Total Assets (100M JPY)	336	3.25e6	2.52e6	6.80e5	1.19e6	1.51e6	5.62e6	7.65e6
Loan Interest Rate (%)	387	1.367	0.688	0.465	0.789	1.309	1.696	4.043
Deposit Interest Rate (%)	489	1.347	2.240	0.003	0.029	0.120	0.956	8.348
1615.T Price	218	157.057	99.039	67.772	103.543	126.243	155.227	647.900
10-Year US T-Bill Yield (%)	772	5.815	2.936	0.624	3.890	5.440	7.527	15.324
CBOE VIX	436	19.475	7.392	10.125	14.114	17.685	23.099	62.669

Notes: Descriptive statistics for all 20 raw variables. Monetary stock variables (M1, M2, M3, Treasury Reserves, BoJ Assets) are reported in levels (JPY or 100M JPY). N differs across variables due to heterogeneous data availability across the sample period.

Table 3.2 – Transformed Variables Descriptive Statistics

Variable	N	Mean	Std	Min	25%	50%	75%	Max
<i>Log-Differenced Variables</i>								
LogDiff – M1 (JPY)	766	0.00729	0.00991	-0.05105	0.00227	0.00549	0.01142	0.08601
LogDiff – M2 (JPY)	745	0.00752	0.00620	-0.01876	0.00240	0.00654	0.01183	0.03045
LogDiff – M3 (JPY)	526	0.00321	0.00310	-0.00526	0.00122	0.00228	0.00484	0.01901
LogDiff – Treasury Reserves (– Gold)	830	0.00834	0.03715	-0.17833	-0.00334	0.00532	0.01561	0.46176
LogDiff – USD-JPY REER CPI (2015=100)	673	0.00011	0.02353	-0.07271	-0.01410	-0.00255	0.01244	0.11027
LogDiff – JPY-USD Spot Exchange Rate	662	-0.00123	0.02588	-0.10523	-0.01482	0.00039	0.01479	0.08066
LogDiff – HICP (SA)	663	2.37013	6.17539	-23.9831	-0.70725	1.12954	4.19913	50.6549
LogDiff – 1615.T Price	217	0.00559	0.07381	-0.35986	-0.03489	0.00635	0.05657	0.24711
LogDiff – BoJ Total Assets (100M JPY)	335	0.00631	0.04301	-0.28671	-0.00729	0.00841	0.02370	0.28005
LogDiff – Call Money/Interbank Rate (%)	390	NaN	NaN	$-\infty$	-0.03944	0.00000	0.02177	$+\infty$
<i>AR(1)-Detrended Variables</i>								
AR(1) – Est. 1-Year Neutral Rate (%)	371	-8.26e-17	1.90e-2	-6.41e-2	2.39e-3	3.68e-3	5.58e-3	7.95e-2
AR(1) – Est. 10-Year Neutral Rate (%)	371	5.16e-16	2.48e-2	-8.37e-2	2.39e-3	4.08e-3	6.79e-3	1.07e-1
AR(1) – Central Gov. Debt (%GDP)	384	-1.62e-13	2.255	-14.283	-0.439	-0.426	-0.411	17.986
AR(1) – Priv. Debt Securities (%GDP)	276	-1.72e-14	1.836	-12.868	-0.221	-0.123	0.106	12.655
AR(1) – Pub. Debt Securities (%GDP)	276	-9.65e-14	4.016	-17.446	-0.740	-0.448	-0.320	32.422
AR(1) – Loan Interest Rate (%)	386	-1.02e-15	0.107	-0.305	-0.066	0.002	0.070	0.288
AR(1) – Deposit Interest Rate (%)	488	4.09e-16	0.143	-0.848	-0.007	0.003	0.013	1.248
AR(1) – 10-Year US T-Bill Yield (%)	771	2.45e-15	0.275	-1.714	-0.139	-0.007	0.147	1.634
AR(1) – CBOE VIX	435	1.32e-14	3.928	-10.442	-1.900	-0.552	1.032	38.133
<i>HP-Filtered Variable</i>								
HPfilter – Real GDP (bn chained 2015 JPY)	384	-2.50e-13	9.39e-3	-6.17e-2	-4.20e-3	4.55e-4	4.83e-3	2.46e-2

Notes: LogDiff denotes first log-difference transformation. AR(1)-detrended variables are residuals from a first-stage AR(1) regression; mean is zero by construction. HPfilter – Real GDP is the cyclical component (output gap) from the Hodrick-Prescott filter. Call Money min/max show $\pm\infty$ due to undefined log-differences when the rate passes through zero.

Table 3.3 – AR(1), Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) Unit Root Testing – Raw Variables

Variable	ADF Test			PP Test		
	Statistic	p-value	Stationary	Statistic	p-value	Stationary
Monetary Aggregates – M1 (JPY)	2.145	0.9988	No	5.499	1.0000	No
Monetary Aggregates – M2 (JPY)	7.421	1.0000	No	15.311	1.0000	No
Monetary Aggregates – M3 (JPY)	6.496	1.0000	No	11.711	1.0000	No
Total Credit – Private Non-Financial (%GDP)	-1.274	0.6410	No	-1.104	0.7137	No
Total Credit – General Government (%GDP)	-3.378	0.0117	Yes	-2.460	0.1256	No
Total Credit – Households & NPISHs (%GDP)	-0.940	0.7747	No	-1.421	0.5723	No
Total Treasury Reserves (– Gold)	-1.598	0.4844	No	-1.615	0.4753	No
10-Year Gov. Bond Yield (%)	-0.735	0.8375	No	-0.282	0.9280	No
Call Money/Interbank Rate (%)	-3.533	0.0072	Yes	-2.928	0.0422	Yes
Est. 1-Year Neutral Interest Rate (%)	-1.544	0.5119	No	-1.480	0.5435	No
Est. 10-Year Neutral Interest Rate (%)	-1.907	0.3289	No	-1.680	0.4415	No
USD-JPY REER CPI-Based (Index 2015=100)	-1.479	0.5437	No	-1.127	0.7041	No
JPY-USD Spot Exchange Rate	-1.250	0.6519	No	-1.199	0.6742	No
HICP (SA)	-0.731	0.8385	No	-0.816	0.8144	No
Real GDP (bn chained 2015 JPY)	-0.479	0.8961	No	-0.892	0.7907	No
Central Government Debt (%GDP)	-1.846	0.3580	No	-2.382	0.1470	No
Domestic Private Debt Securities (%GDP)	-1.449	0.5585	No	-1.449	0.5584	No
Domestic Public Debt Securities (%GDP)	-2.172	0.2165	No	-2.281	0.1780	No
BoJ Total Assets (100M JPY)	-0.280	0.9283	No	5.104	1.0000	No
Loan Interest Rate (%)	-2.635	0.0860	No	-1.787	0.3869	No
Deposit Interest Rate (%)	-3.939	0.0018	Yes	-2.433	0.1326	No
1615.T Price	-2.483	0.1197	No	-2.532	0.1079	No
10-Year US T-Bill Yield (%)	-2.388	0.1450	No	-1.955	0.3066	No
CBOE VIX	-2.560	0.1015	No	-2.380	0.1474	No

Notes: Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) tests are conducted with automatic lag selection (AIC) and a constant term. The null hypothesis (H_0) is the presence of a unit root; “Stationary” indicates rejection of H_0 at the 5% significance level ($p < 0.05$). ADF corrects for serial correlation via lagged differences; PP uses a non-parametric Newey–West long-run variance correction. Statistics are rounded to three decimal places.

Table 3.4 – AR(1), Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) Unit Root Testing – Transformed Variables

Variable	AR(1)	ADF Test			PP Test		
		Statistic	p-value	Stat.	Statistic	p-value	Stat.
<i>Log-Differenced Variables</i>							
LogDiff – M1 (JPY)	0.345	-2.444	0.1296	No	-7.836	6.12e-12	Yes
LogDiff – M2 (JPY)	0.167	-8.190	7.73e-13	Yes	-8.292	4.23e-13	Yes
LogDiff – M3 (JPY)	0.267	-7.423	6.68e-11	Yes	-7.608	2.29e-11	Yes
LogDiff – Treasury Reserves (– Gold)	-0.012	-4.323	4.04e-04	Yes	-9.845	4.64e-17	Yes
LogDiff – USD-JPY REER CPI (2015=100)	0.360	-6.642	5.39e-09	Yes	-6.364	2.43e-08	Yes
LogDiff – JPY-USD Spot Exchange Rate	0.317	-6.897	1.31e-09	Yes	-6.926	1.11e-09	Yes
LogDiff – HICP (SA)	0.303	-7.072	4.90e-10	Yes	-7.211	2.23e-10	Yes
LogDiff – 1615.T Price	0.084	-8.614	6.36e-14	Yes	-8.633	5.69e-14	Yes
LogDiff – BoJ Total Assets (100M JPY)	-0.234	-1.240	0.6563	No	-12.010	3.20e-22	Yes
LogDiff – Call Money/Interbank Rate (%)	0.164	-6.230	4.97e-08	Yes	-6.218	5.30e-08	Yes
<i>AR(1)-Detrended Variables</i>							
AR(1) – Total Credit – Gen. Gov. (%GDP)	-0.192	-3.551	6.77e-03	Yes	-11.790	9.87e-22	Yes
AR(1) – Total Credit – HH & NPISHs (%GDP)	0.001	-3.680	4.40e-03	Yes	-10.636	5.08e-19	Yes
AR(1) – Total Credit – Priv. Non-Fin. (%GDP)	-0.003	-3.522	7.44e-03	Yes	-10.264	4.17e-18	Yes
AR(1) – 10-Year Gov. Bond Yield (%)	0.012	-8.552	9.16e-14	Yes	-10.519	9.81e-19	Yes
AR(1) – Call Money/Interbank Rate (%)	0.451	-5.842	3.76e-07	Yes	-5.669	9.00e-07	Yes
AR(1) – Est. 1-Year Neutral Rate (%)	-0.039	-2.197	0.2074	No	-11.350	1.00e-20	Yes
AR(1) – Est. 10-Year Neutral Rate (%)	-0.026	-2.157	0.2222	No	-11.330	1.12e-20	Yes
AR(1) – Central Gov. Debt (%GDP)	-0.047	-2.534	0.1075	No	-10.902	1.15e-19	Yes
AR(1) – Priv. Debt Securities (%GDP)	-0.018	-9.881	3.78e-17	Yes	-9.988	2.03e-17	Yes
AR(1) – Pub. Debt Securities (%GDP)	-0.001	-9.703	1.06e-16	Yes	-9.707	1.04e-16	Yes
AR(1) – Loan Interest Rate (%)	-0.490	-8.935	9.60e-15	Yes	-31.693	≈0	Yes
AR(1) – Deposit Interest Rate (%)	-0.243	-2.839	0.0529	No	-12.106	1.97e-22	Yes
AR(1) – 10-Year US T-Bill Yield (%)	0.221	-7.658	1.72e-11	Yes	-7.412	7.10e-11	Yes
AR(1) – CBOE VIX	0.247	-7.445	5.88e-11	Yes	-7.298	1.36e-10	Yes
<i>HP-Filtered Variable</i>							
HPfilter – Real GDP (bn chained 2015 JPY)	0.792	-3.862	2.33e-03	Yes	-3.181	2.11e-02	Yes

Notes: AR(1) coefficient is the first-order autocorrelation of the transformed series. ADF and PP tests use automatic lag selection (AIC) with a constant term; “Stat.” indicates rejection of H_0 (unit root present) at the 5% significance level ($p < 0.05$). Conflicts between ADF and PP outcomes reflect differences in serial correlation correction: ADF uses parametric lagged differences; PP applies a non-parametric Newey–West long-run variance correction. For AR(1)-detrended variables, stationarity was the transformation objective; residual non-rejection (ADF) in a handful of cases (Est. Neutral Rates, Central Gov. Debt, Deposit Rate) is superseded by strong PP rejection. The PP statistic for AR(1) – Loan Interest Rate (%) is -31.693 ($p \approx 0$). p-values reported in scientific notation where $|p| < 0.01$.

Table 4.1 – FC-FAVAR Marginal Factor PCA

Factor	Marginal Variance (%)	Cumulative Variance (%)	Retained
F1	51.623	51.623	
F2	24.244	75.867	
F3	12.930	88.797	
F4	7.126	95.923	
F5	3.071	98.994	

Notes: Principal Component Analysis (PCA) applied to the 20 transformed FAVAR variables to extract 5 latent factors (F1–F5). Marginal variance denotes the share of total variance explained by each individual factor; cumulative variance is the running sum.

Table 4.2 – FC-FAVAR Lag Order Selection (AIC, BIC, FPE, HQIC)

Lag	AIC	BIC	FPE	HQIC
0	-13.960	-13.920	8.675×10^{-7}	-13.940
1	-14.980	-14.830	3.134×10^{-7}	-14.920
2	-15.370	-15.110	2.107×10^{-7}	-15.270
3	-15.500	-15.130*	1.858×10^{-7}	-15.350*
4	-15.510*	-15.030	1.837×10^{-7}*	-15.320
5	-15.480	-14.880	1.898×10^{-7}	-15.240
6	-15.510	-14.800	1.845×10^{-7}	-15.220
7	-15.480	-14.660	1.902×10^{-7}	-15.150
8	-15.480	-14.550	1.902×10^{-7}	-15.100
9	-15.500	-14.460	1.863×10^{-7}	-15.080
10	-15.490	-14.340	1.875×10^{-7}	-15.030

Notes: Lag order selection for the FC-FAVAR based on four information criteria: Akaike (AIC), Bayesian/Schwarz (BIC), Final Prediction Error (FPE), and Hannan–Quinn (HQIC). Asterisks (*) and bold entries indicate the lag minimising each criterion.

Table 4.3 – Functional Coefficient z_t Identified Regimes

Regime	Value
z_1	-0.84730
z_2	-0.14590
z_3	0.00000
z_4	0.01047

Notes: Identified values of the functional coefficient transition variable z_t defining the four estimated regimes of the FC-FAVAR.

Table 4.4 – FC-FAVAR Local VAR Diagonal AR(1) Coefficients and Stability Check

Regime (z_0)	$\hat{\phi}_{11}$	$\hat{\phi}_{22}$	$\hat{\phi}_{33}$	$\hat{\phi}_{44}$
$z_1 = -0.847$	0.581	1.161	0.404	-0.256
$z_2 = -0.146$	-0.207	-0.593	0.517	0.148
$z_3 = 0.000$	0.231	-0.503	0.242	0.140
$z_4 = 0.010$	0.230	-0.420	0.237	0.146

Stability – Eigenvalues of Average A_1 Matrix

$|\lambda_1| = 0.411, \quad |\lambda_{2,3}| = 0.231, \quad |\lambda_{4,5}| = 0.126, \quad |\lambda_{6,7}| = 0.097, \quad |\lambda_8| = 0.055$

Notes: $\hat{\phi}_{kk}$ denotes the diagonal AR(1) coefficient of the k -th factor in the locally-weighted VAR estimated at regime z_0 . Local weights are computed via Gaussian kernel rescaling with bandwidth $h = 0.1 \cdot \hat{\sigma}_Z$, prioritising observations in the neighbourhood of each z_0 . Coefficients are estimated by weighted OLS using the Moore–Penrose pseudoinverse. The stability check reports the modulus of eigenvalues of the average first-lag companion matrix $\bar{A}_1 = \frac{1}{4} \sum_{i=1}^4 A_1(z_i)$; all eigenvalues satisfy $|\lambda| < 1$, confirming mean stationarity of the average system.

Table 5.1 – FC-FAVAR Local VAR Estimated Coefficient Matrices by Regime

Regime 1 $z_0 = -0.847$									
		M3	CPNF	CM	F1	F2	F3	F4	F5
$\hat{\Phi}_1$									
M3		0.5595	-44.7507	-0.0358	-143.413	46.757	-483.948	-177.086	-9.128
CPNF		0.0058	0.4687	0.0156	0.316	4.147	1.119	1.892	-0.100
CM		-0.0019	-2.5330	0.3879	-32.109	2.620	0.934	-0.444	-2.786
F1		0.0000	0.0403	0.0001	-0.179	0.025	-0.031	0.080	0.009
F2		0.0000	0.0061	0.0001	0.078	0.054	0.020	-0.014	0.016
F3		0.0003	0.0216	0.0010	0.342	0.216	0.066	0.118	-0.013
F4		-0.0020	-0.1813	-0.0056	-0.426	-1.683	-0.319	-0.720	0.039
F5		-0.0050	-0.4345	-0.0143	-0.471	-4.187	-0.937	-1.656	0.056
$\hat{\Phi}_2$									
M3		0.0455	-39.7214	0.1278	-437.192	14.847	166.061	31.537	-48.792
CPNF		0.0154	2.6430	0.0148	23.151	15.539	-0.783	4.129	1.571
CM		0.0004	-1.0019	0.6178	7.124	-5.227	-0.816	-2.849	0.029
F1		0.0002	0.0381	0.0000	0.125	0.160	-0.085	0.040	0.023
F2		0.0001	0.0140	0.0001	-0.131	0.033	-0.009	0.046	-0.004
F3		0.0009	0.1292	0.0011	3.311	0.797	-0.029	0.220	0.089
F4		-0.0054	-0.9426	-0.0054	-8.602	-6.012	0.261	-1.591	-0.521
F5		-0.0136	-2.3575	-0.0138	-20.201	-14.489	0.782	-3.707	-1.388
$\hat{\Phi}_3$									
M3		0.0421	42.8795	0.0050	612.847	-97.768	-7.542	92.024	14.080
CPNF		-0.0025	5.0100	-0.0018	-12.243	36.687	3.872	11.480	1.041
CM		0.0023	2.2486	-0.4824	38.920	-3.694	-1.217	1.054	2.254
F1		0.0000	0.0207	-0.0001	-0.293	0.373	0.104	0.150	-0.033
F2		-0.0001	-0.0564	-0.0000	0.116	0.377	-0.045	-0.042	-0.047
F3		-0.0001	0.7413	0.0001	-1.006	1.864	0.342	1.246	0.319
F4		0.0008	-1.8012	0.0006	4.526	-13.505	-1.231	-4.037	-0.393
F5		0.0023	-4.1596	0.0014	11.799	-33.689	-2.789	-9.813	-0.708

Table 5.1 (cont.) – FC-FAVAR Local VAR Estimated Coefficient Matrices by Regime

Regime 2 $z_0 = -0.146$									
		M3	CPNF	CM	F1	F2	F3	F4	F5
$\hat{\Phi}_1$									
M3		-0.2395	-82.3757	-58.858	746.857	-275.755	-113.851	174.289	-155.803
CPNF		-0.0066	-0.0540	0.350	-5.270	9.038	-1.447	-0.690	0.155
CM		0.0006	-0.2620	0.376	2.871	-1.680	-0.157	0.153	-0.376
F1		-0.0001	0.0351	0.001	0.172	0.090	-0.029	-0.005	0.041
F2		0.0001	-0.0230	-0.008	-0.120	0.192	-0.025	0.005	-0.026
F3		-0.0003	-0.0208	0.027	-0.221	0.357	-0.144	-0.069	-0.005
F4		0.0027	0.0030	-0.128	2.295	-3.527	0.488	0.248	-0.077
F5		0.0059	0.1371	-0.297	3.970	-8.288	1.390	0.655	-0.060
$\hat{\Phi}_2$									
M3		0.0479	-86.1256	32.269	-601.115	-641.534	-16.942	170.726	-181.596
CPNF		0.0055	-2.2619	-0.723	7.044	5.576	0.786	-3.712	-0.817
CM		-0.0000	-0.3027	0.050	-0.657	-0.283	0.256	0.647	-0.598
F1		0.0001	-0.0304	-0.009	-0.281	-0.138	-0.036	-0.014	-0.034
F2		-0.0001	0.0005	0.011	-0.087	-0.133	-0.129	-0.009	-0.009
F3		0.0004	-0.1399	-0.062	0.359	0.325	0.029	-0.259	-0.041
F4		-0.0018	0.7678	0.220	-2.575	-2.123	-0.310	1.323	0.246
F5		-0.0047	2.0119	0.690	-5.775	-5.278	-0.616	3.445	0.673
$\hat{\Phi}_3$									
M3		0.2769	79.1491	-3.860	-488.434	570.047	-19.285	-68.360	114.724
CPNF		0.0008	1.0392	-0.995	6.198	-23.207	6.713	11.894	-3.800
CM		-0.0009	0.1360	0.196	-3.548	1.782	-0.647	-0.989	0.516
F1		-0.0000	-0.0057	-0.010	0.013	-0.314	0.116	0.033	-0.017
F2		0.0003	0.0122	-0.006	0.085	-0.107	0.132	0.079	-0.015
F3		0.0001	-0.0885	-0.081	0.264	-1.525	0.526	0.828	-0.426
F4		-0.0001	0.0059	0.349	-1.566	8.428	-2.007	-3.843	1.647
F5		-0.0004	-0.6706	0.907	-5.307	21.065	-5.802	-10.539	3.663

Table 5.1 (cont.) – FC-FAVAR Local VAR Estimated Coefficient Matrices by Regime

Regime 3 $z_0 = 0.000$		M3	CPNF	CM	F1	F2	F3	F4	F5
$\hat{\Phi}_1$									
M3		0.2737	-95.6237	-9.501	262.688	-28.812	-80.842	-34.640	-88.189
CPNF		0.0002	-0.2490	-0.314	-10.526	-0.195	-2.545	-0.170	-0.413
CM		-0.0001	-0.1071	0.253	2.032	0.976	0.248	-0.082	-0.048
F1		-0.0000	0.0351	-0.016	0.276	-0.152	-0.071	0.020	0.027
F2		-0.0001	0.0046	0.005	-0.169	0.166	0.019	0.033	-0.004
F3		0.0001	0.0014	-0.020	-0.493	-0.005	-0.239	0.008	-0.017
F4		0.0001	0.0836	0.079	4.099	0.093	0.892	0.079	0.139
F5		-0.0002	0.1537	0.266	9.416	0.252	2.522	0.145	0.310
$\hat{\Phi}_2$									
M3		0.1140	52.2116	-0.293	-114.014	136.784	74.850	40.927	46.301
CPNF		0.0027	0.5897	0.724	5.685	1.991	-0.354	0.480	0.471
CM		-0.0004	-0.2009	0.261	-0.750	0.092	0.556	0.280	-0.323
F1		0.0000	-0.0045	0.009	-0.176	-0.071	0.002	-0.003	-0.005
F2		0.0000	-0.0088	-0.000	-0.117	0.016	-0.054	0.000	-0.016
F3		0.0002	0.0694	0.035	0.234	0.066	-0.062	0.035	0.058
F4		-0.0009	-0.1901	-0.266	-2.377	-0.876	-0.087	-0.167	-0.167
F5		-0.0023	-0.5961	-0.673	-4.455	-1.750	0.404	-0.463	-0.472
$\hat{\Phi}_3$									
M3		0.1804	-152.191	-27.676	-99.696	320.470	150.188	88.867	-189.671
CPNF		-0.0039	-4.7952	-2.068	-2.495	-1.499	2.741	1.031	-5.514
CM		0.0001	0.0320	0.067	-0.090	0.096	0.006	0.137	-0.028
F1		-0.0000	0.0049	-0.024	0.004	-0.036	0.211	-0.003	0.021
F2		-0.0000	0.0074	-0.019	0.337	0.046	0.046	-0.034	0.025
F3		-0.0002	-0.2882	-0.148	-0.261	-0.388	0.252	0.296	-0.420
F4		0.0014	1.7697	0.735	1.102	0.062	-0.398	-0.389	2.078
F5		0.0033	4.5979	1.920	2.533	1.437	-2.159	-0.708	5.215

Table 5.1 (cont.) – FC-FAVAR Local VAR Estimated Coefficient Matrices by Regime

Regime 4 $z_0 = 0.010$		M3	CPNF	CM	F1	F2	F3	F4	F5
$\hat{\Phi}_1$									
M3		0.2711	-89.5881	-7.857	278.389	-16.957	-75.243	-29.475	-83.108
CPNF		0.0002	-0.1884	-0.287	-10.659	-0.228	-2.710	-0.168	-0.359
CM		-0.0001	-0.0965	0.243	2.021	0.992	0.247	-0.090	-0.031
F1		-0.0000	0.0373	-0.016	0.275	-0.151	-0.075	0.019	0.030
F2		-0.0001	0.0034	0.005	-0.168	0.164	0.020	0.032	-0.005
F3		0.0001	0.0039	-0.020	-0.495	-0.005	-0.243	0.008	-0.015
F4		0.0001	0.0560	0.068	4.170	0.101	0.957	0.079	0.113
F5		-0.0002	0.0937	0.247	9.560	0.278	2.688	0.149	0.254
$\hat{\Phi}_2$									
M3		0.1178	49.4536	2.581	-147.629	147.102	73.822	28.958	47.880
CPNF		0.0026	0.5422	0.763	6.069	1.917	-0.438	0.398	0.449
CM		-0.0004	-0.1890	0.272	-0.791	0.088	0.552	0.279	-0.311
F1		0.0000	-0.0037	0.008	-0.173	-0.068	0.005	-0.002	-0.004
F2		0.0000	-0.0078	0.000	-0.108	0.021	-0.055	0.003	-0.016
F3		0.0002	0.0666	0.037	0.267	0.063	-0.072	0.031	0.057
F4		-0.0009	-0.1752	-0.283	-2.531	-0.860	-0.061	-0.137	-0.162
F5		-0.0022	-0.5550	-0.708	-4.816	-1.674	0.482	-0.388	-0.455
$\hat{\Phi}_3$									
M3		0.1767	-152.656	-28.450	-75.765	299.160	141.864	89.749	-191.074
CPNF		-0.0038	-4.8594	-2.056	-2.532	-1.207	2.514	0.857	-5.526
CM		0.0001	0.0403	0.072	-0.063	0.067	0.020	0.163	-0.030
F1		-0.0000	0.0051	-0.023	0.012	-0.034	0.201	-0.004	0.020
F2		-0.0001	0.0036	-0.019	0.339	0.049	0.041	-0.039	0.023
F3		-0.0002	-0.2971	-0.148	-0.260	-0.387	0.239	0.287	-0.426
F4		0.0014	1.7927	0.730	1.085	-0.056	-0.314	-0.321	2.080
F5		0.0033	4.6490	1.908	2.547	1.175	-1.969	-0.561	5.221

Notes: Estimated local VAR coefficient matrices $\hat{\Phi}_q(z_0)$ for lags $q \in \{1, 2, 3\}$ at each of the four regime centroids $z_0 \in \{-0.847, -0.146, 0.000, 0.010\}$. Estimation uses Gaussian kernel-weighted OLS with bandwidth $h = 0.1\hat{\sigma}_Z$. Column and row labels are abbreviated: **M3** = LogDiff-M3 (JPY); **CPNF** = AR(1)-detrended Total Credit – Private Non-Financial (%GDP); **CM** = LogDiff-Call Money/Interbank Rate (%); **F1–F5** = PCA latent factors. Lag order $p = 4$ selected by AIC (Table 4.2); only lags 1–3 shown as $\hat{\Phi}_4$ is available on request. Coefficients are rounded to three decimal places.

Table 5.2 – FC-FAVAR-Derived Granger Causality: M3 $\longleftrightarrow$ Credit by Regime

Direction	<i>F</i> -stat	df	p-value	Decision
Regime 1 $z_0 = -0.847$				
M3 $\rightarrow$ Credit	-3.741	(3, 273)	1.0000	Fail to reject H_0
Credit $\rightarrow$ M3	-47.979	(3, 273)	1.0000	Fail to reject H_0
<i>No Granger causality detected</i>				
Regime 2 $z_0 = -0.146$				
M3 $\rightarrow$ Credit	-1.109	(3, 273)	1.0000	Fail to reject H_0
Credit $\rightarrow$ M3	1.269	(3, 273)	0.2852	Fail to reject H_0
<i>No Granger causality detected</i>				
Regime 3 $z_0 = 0.000$				
M3 $\rightarrow$ Credit	-6.251	(3, 273)	1.0000	Fail to reject H_0
Credit $\rightarrow$ M3	2.670	(3, 273)	0.0479**	Reject H_0
<i>Unidirectional: Credit $\rightarrow$ M3</i>				
Regime 4 $z_0 = 0.010$				
M3 $\rightarrow$ Credit	-6.222	(3, 273)	1.0000	Fail to reject H_0
Credit $\rightarrow$ M3	2.723	(3, 273)	0.0447**	Reject H_0
<i>Unidirectional: Credit $\rightarrow$ M3</i>				

Notes: FC-FAVAR-Derived Granger causality tests conducted within the FC-FAVAR local VAR at each regime centre z_0 , with lag order $p = 3$ and $F(3, 273)$ distributed test statistics. H_0 : lagged values of X do not Granger-cause Y . ** denotes rejection at the 5% significance level.

Table 5.3a – Global FC-FAVAR Estimation Results

Regressor	M3 $R^2 = 0.639$		CPNF $R^2 = 0.329$		CM $R^2 = 0.630$		F1 $R^2 = 0.199$	
	Coef	p	Coef	p	Coef	p	Coef	p
L1.M3	0.4089	0.000***	-0.0024	0.201	0.0005	0.047**	-0.0001	0.024**
L1.CPNF	8.7756	0.870	-0.0264	0.987	0.0058	0.977	0.0706	0.000***
L1.CM	-1.5130	0.930	-0.4698	0.383	0.2452	0.000***	-0.0098	0.122
L1.F1	30.184	0.891	-1.7209	0.803	0.9722	0.246	0.0920	0.256
L1.F2	94.385	0.550	1.1093	0.823	-0.3238	0.590	-0.0618	0.288
L1.F3	-303.138	0.008***	-0.6864	0.847	0.1316	0.760	-0.0339	0.416
L1.F4	70.957	0.334	0.4242	0.854	-0.1059	0.704	0.0726	0.007***
L1.F5	-34.313	0.499	-0.1857	0.907	0.0500	0.796	0.0464	0.013**
L2.M3	0.0938	0.141	0.0036	0.041**	-0.0004	0.054*	0.0001	0.000***
L2.CPNF	-61.588	0.275	0.0135	0.993	-0.2298	0.254	0.0003	0.989
L2.CM	-1.9917	0.912	-0.0308	0.951	0.3547	0.000***	-0.0008	0.895
L2.F1	-134.481	0.562	-0.0104	0.999	0.0208	0.980	-0.1405	0.079*
L2.F2	-87.325	0.600	5.2787	0.253	-0.3880	0.514	0.0165	0.774
L2.F3	194.336	0.104	-2.1194	0.521	0.3012	0.480	-0.0594	0.149
L2.F4	-78.757	0.308	0.9095	0.671	-0.0660	0.811	0.0035	0.895
L2.F5	-23.651	0.658	-0.4010	0.786	-0.2294	0.230	-0.0048	0.793
L3.M3	0.1076	0.064*	-0.0027	0.120	-0.0003	0.182	0.0000	0.206
L3.CPNF	-74.454	0.148	-2.9289	0.060*	0.2417	0.236	-0.0158	0.428
L3.CM	-21.018	0.202	-1.3372	0.008***	0.1442	0.028**	-0.0136	0.033**
L3.F1	9.1036	0.966	1.3997	0.826	0.0477	0.955	0.0004	0.996
L3.F2	-47.528	0.754	3.9962	0.383	-0.1672	0.781	-0.0250	0.670
L3.F3	10.506	0.923	1.7111	0.602	-0.6224	0.150	0.1339	0.002***
L3.F4	40.381	0.566	-1.1334	0.594	0.2909	0.298	-0.0267	0.327
L3.F5	-96.057	0.049**	-2.6868	0.068*	0.1027	0.595	0.0012	0.951
L4.M3	0.0091	0.857	0.0028	0.118	-0.0001	0.577	0.0000	0.210
L4.CPNF	64.806	0.149	0.5253	0.739	-0.3074	0.132	-0.0005	0.981
L4.CM	4.6612	0.745	0.1893	0.708	0.1824	0.006***	0.0035	0.588
L4.F1	335.816	0.069*	-2.3809	0.713	0.3109	0.711	-0.0331	0.688
L4.F2	104.384	0.430	-0.9992	0.830	-0.3850	0.522	0.0276	0.641
L4.F3	-34.504	0.716	-2.5664	0.441	0.4916	0.255	0.0114	0.789
L4.F4	31.741	0.605	0.7630	0.724	-0.1417	0.612	0.0266	0.334
L4.F5	60.533	0.154	0.0747	0.960	-0.2582	0.182	-0.0106	0.577

Notes: First four equations (M3, CPNF, CM, F1). **M3** = LogDiff-M3 (JPY); **CPNF** = AR(1)-detrended Total Credit – Private Non-Financial (%GDP); **CM** = LogDiff-Call Money (%); **F1–F5** = PCA latent factors. Lag order $p = 4$ selected by AIC. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5.3a (cont.) – Global FC-FAVAR Estimation Results

Regressor	F2		F3		F4		F5	
	$R^2 = 0.134$		$R^2 = 0.173$		$R^2 = 0.368$		$R^2 = 0.301$	
	Coef	p	Coef	p	Coef	p	Coef	p
L1.M3	−0.0000	0.191	−0.0001	0.341	0.0010	0.150	0.0023	0.183
L1.CPNF	−0.0033	0.896	0.0078	0.939	0.0121	0.985	0.0163	0.992
L1.CM	−0.0016	0.839	−0.0234	0.475	0.1536	0.447	0.4323	0.382
L1.F1	−0.0676	0.508	0.0973	0.817	0.7090	0.784	1.7900	0.778
L1.F2	0.2049	0.005***	0.0335	0.911	−0.5246	0.778	−1.2264	0.788
L1.F3	−0.0300	0.568	−0.1038	0.631	0.2416	0.856	0.6449	0.843
L1.F4	0.0180	0.597	0.0034	0.981	−0.1799	0.835	−0.3534	0.867
L1.F5	−0.0093	0.692	0.0059	0.952	0.0795	0.894	0.1474	0.920
L2.M3	0.0000	0.177	0.0003	0.013**	−0.0011	0.082*	−0.0030	0.062*
L2.CPNF	−0.0060	0.804	0.0023	0.981	0.0181	0.975	−0.0567	0.968
L2.CM	0.0001	0.994	−0.0008	0.979	0.0084	0.964	0.0216	0.962
L2.F1	−0.0863	0.385	0.4327	0.278	−0.1172	0.961	0.8113	0.890
L2.F2	0.0127	0.859	0.2555	0.372	−2.0687	0.228	−4.7688	0.259
L2.F3	−0.0146	0.776	−0.1770	0.388	0.6420	0.601	2.0164	0.506
L2.F4	0.0284	0.391	0.0423	0.750	−0.3460	0.663	−0.7826	0.690
L2.F5	−0.0205	0.370	−0.0182	0.843	0.1649	0.764	0.3104	0.819
L3.M3	−0.0000	0.183	−0.0001	0.251	0.0010	0.136	0.0024	0.131
L3.CPNF	−0.0091	0.710	−0.1440	0.140	1.1547	0.046**	3.0707	0.031**
L3.CM	−0.0171	0.031**	−0.0904	0.004***	0.4880	0.009***	1.2355	0.007***
L3.F1	0.2511	0.013**	−0.1189	0.767	−0.5091	0.830	−1.2959	0.824
L3.F2	0.0629	0.384	0.0791	0.783	−1.7173	0.312	−3.6346	0.386
L3.F3	−0.0110	0.832	0.1777	0.389	−0.3214	0.792	−1.2435	0.679
L3.F4	−0.0578	0.085*	0.2506	0.061*	0.5741	0.466	1.3234	0.496
L3.F5	0.0127	0.585	−0.2520	0.007***	1.0316	0.059*	2.7975	0.038**
L4.M3	0.0000	0.085*	0.0003	0.015**	−0.0011	0.110	−0.0026	0.108
L4.CPNF	0.0007	0.979	0.0610	0.539	−0.1452	0.804	−0.4904	0.735
L4.CM	−0.0075	0.360	0.0301	0.345	−0.0675	0.719	−0.2036	0.661
L4.F1	0.0956	0.363	−0.1400	0.732	0.9205	0.702	1.8656	0.754
L4.F2	0.0104	0.890	−0.0313	0.915	0.3684	0.831	1.0712	0.802
L4.F3	0.0283	0.601	−0.2069	0.325	0.8852	0.475	2.4451	0.425
L4.F4	0.0212	0.544	0.0786	0.563	−0.2458	0.759	−0.6424	0.746
L4.F5	−0.0056	0.818	0.0193	0.838	0.0060	0.991	−0.0962	0.944

Notes: Last four equations (F2–F5). Abbreviations and significance levels as per Table 5.3a.

Table 5.4 – Global FC-FAVAR Impulse Response Functions: M3 ↔ Credit

Horizon	M3 → Credit	Credit → M3
$h = 0$	0.0976	0.0000
$h = 1$	0.8834	134.3344
$h = 2$	1.0650	−95.940
$h = 3$	0.0936	−133.931
$h = 4$	2.5922	−111.341
$h = 5$	0.8780	20.035
$h = 6$	−0.8998	−86.140
$h = 7$	0.3655	48.627
$h = 8$	0.3266	3.251
$h = 9$	0.1461	−63.222
$h = 10$	0.2130	21.269
$h = 11$	0.3461	30.378
$h = 12$	0.0839	−25.666
$h = 13$	0.2420	27.726
$h = 14$	0.2719	18.599
$h = 15$	0.1267	−10.621
$h = 16$	0.1767	16.937
$h = 17$	0.2441	15.704
$h = 18$	0.0913	−3.087
$h = 19$	0.1473	14.018
$h = 20$	0.1772	10.947
$h = 21$	0.0900	1.147
$h = 22$	0.1144	10.024
$h = 23$	0.1367	8.980
$h = 24$	0.0763	3.072

Notes: Impulse response functions from the global (pooled, no-regime) FC-FAVAR over a 24-period horizon.

Table 5.5 – Three-Agent IRL Model: Transition Dynamics Equations Parameters

Variable	Coef	Std Err	<i>t</i>	p-value	[0.025	0.975]
<i>Output Gap</i> _{<i>t</i>+1} $R^2 = 0.5613$						
Constant	−0.0001	0.000	−0.260	0.795	−0.001	0.001
Output Gap _{<i>t</i>}	0.7518	0.041	18.264	0.000***	0.671	0.833
Credit (PNF) _{<i>t</i>}	−0.0006	0.001	−1.217	0.225	−0.002	0.000
Capital Stress _{<i>t</i>}	−0.0003	9.3×10^{-5}	−3.670	0.000***	−0.001	−0.000
<i>Inflation</i> _{<i>t</i>+1} $R^2 = 0.0418$						
Constant	0.0620	0.164	0.377	0.706	−0.262	0.386
Inflation _{<i>t</i>}	0.1633	0.059	2.764	0.006***	0.047	0.280
Output Gap _{<i>t</i>+1}	26.9646	15.817	1.705	0.089*	−4.170	58.100
<i>Default Proxy</i> _{<i>t</i>+1} $R^2 = 0.0455$						
Constant	0.4613	0.054	8.476	0.000***	0.354	0.568
Default Proxy _{<i>t</i>}	−0.2198	0.087	−2.533	0.012**	−0.391	−0.049
Neg. Output Gap _{<i>t</i>+1}	−12.3928	6.214	−1.994	0.047**	−24.624	−0.161
Credit (PNF) _{<i>t</i>}	−0.0544	0.060	−0.913	0.362	−0.172	0.063
<i>Capital Stress</i> _{<i>t</i>+1} $R^2 = 0.0393$						
Constant	−0.1435	0.301	−0.477	0.634	−0.736	0.449
Capital Stress _{<i>t</i>}	0.0246	0.059	0.419	0.675	−0.091	0.140
Default Proxy _{<i>t</i>+1}	1.2051	0.359	3.353	0.001***	0.498	1.913
NIM _{<i>t</i>}	−2.2835	2.497	−0.915	0.361	−7.198	2.631
<i>Household Debt Proxy</i> _{<i>t</i>+1} $R^2 = 0.9708$						
Constant	0.0015	0.025	0.060	0.953	−0.049	0.052
HH Debt Proxy _{<i>t</i>}	0.9892	0.011	93.099	0.000***	0.968	1.010
Credit (HH) _{<i>t</i>}	0.0232	0.078	0.295	0.768	−0.131	0.178
Real Rate _{<i>t</i>}	−0.0064	0.008	−0.758	0.449	−0.023	0.010

Notes: OLS estimates of the five transition equations of the Three-Agent IRL model (Central Bank, Commercial Bank, Household). Each panel reports one state variable equation estimated at lag 1. R^2 is reported per equation. Standard errors are heteroskedasticity-consistent. Confidence intervals at the 95% level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5.6 – Three-Agent IRL Model: Inferred Reward Functions

Variable	Coef	Std Err	t	p-value	[0.025	0.975]
<i>BoJ Policy Rate</i> $R^2 = 0.0309$						
Constant	6.18×10^{-5}	0.002	0.034	0.973	-0.003	0.004
Inflation Gap	0.0008	0.002	0.435	0.664	-0.003	0.004
Output Gap	-0.0006	0.002	-0.300	0.764	-0.004	0.003
Capital Stress	-0.0109	0.008	-1.412	0.159	-0.026	0.004
Gov. Debt	-0.0008	0.002	-0.425	0.671	-0.004	0.003
FX	-0.0016	0.002	-0.850	0.396	-0.005	0.002
VIX	0.0062	0.008	0.819	0.413	-0.009	0.021
<i>BoJ Quantitative Easing (QE)</i> $R^2 = 0.0534$						
Constant	0.0079	0.003	2.932	0.004***	0.003	0.013
Inflation Gap	0.0013	0.003	0.464	0.643	-0.004	0.007
Output Gap	0.0008	0.003	0.273	0.785	-0.005	0.007
Capital Stress	-0.0323	0.012	-2.809	0.005***	-0.055	-0.010
Gov. Debt	-0.0051	0.003	-1.848	0.066*	-0.010	0.000
FX	-0.0026	0.003	-0.925	0.356	-0.008	0.003
VIX	0.0330	0.011	2.905	0.004***	0.011	0.055
<i>Commercial Bank</i> $R^2 = 0.6628$						
Constant	-0.1469	0.041	-3.566	0.000***	-0.228	-0.066
NIM	-0.0222	0.042	-0.522	0.602	-0.106	0.061
Capital Stress	-0.1248	0.042	-2.940	0.004***	-0.208	-0.041
Default Proxy	-0.6705	0.045	-14.767	0.000***	-0.760	-0.581
Call Rate	0.0364	0.042	0.863	0.389	-0.047	0.120
Credit (HH)	0.4329	0.045	9.682	0.000***	0.345	0.521
<i>Household</i> $R^2 = 0.1510$						
Constant	-0.0372	0.017	-2.145	0.033**	-0.071	-0.003
Real Rate	-0.6837	2.000	-0.342	0.733	-4.621	3.253
Output Gap	-0.0943	0.018	-5.380	0.000***	-0.129	-0.060
Inflation	-0.6703	2.001	-0.335	0.738	-4.609	3.268
Deposit Rate	0.0087	0.020	0.436	0.663	-0.031	0.048
HH Debt Proxy	0.0723	0.017	4.146	0.000***	0.038	0.107

Notes: OLS estimates of the inferred reward functions $\hat{\Omega}$ for each agent in the Three-Agent IRL model. Each panel reports the regression of the agent's observed action on its state space variables. R^2 per equation. Standard errors are heteroskedasticity-consistent; 95% confidence intervals reported. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5.6 – Three-Agent IRL Model: Inferred Reward Weights $\hat{\Omega}$

State Variable	$\hat{\omega}$
<i>Bank of Japan – Policy Rate</i> $\hat{\Omega}^{BoJ,rate}$	
Inflation Gap	0.0008
Output Gap	-0.0006
Capital Stress	-0.0109
Gov. Debt	-0.0008
<i>Bank of Japan – Quantitative Easing</i> $\hat{\Omega}^{BoJ,QE}$	
Inflation Gap	0.0013
Output Gap	0.0008
Capital Stress	-0.0323
Gov. Debt	-0.0051
<i>Commercial Bank</i> $\hat{\Omega}^{bank}$	
NIM	-0.0222
Capital Stress	-0.1248
Default Proxy	-0.6705
Credit (HH)	0.4329
<i>Household</i> $\hat{\Omega}^{HH}$	
Real Rate	-0.6837
Output Gap	-0.0943
Inflation	-0.6703
Deposit Rate	0.0087
HH Debt Proxy	0.0723

Notes: Inferred reward weights $\hat{\omega}$ from the Three-Agent IRL model for each agent's action equation, excluding intercepts. $\hat{\Omega}^{BoJ,rate}$ and $\hat{\Omega}^{BoJ,QE}$ represent the BoJ's implicit preference weights over state variables for the policy rate and balance sheet (QE) instruments respectively. $\hat{\Omega}^{bank}$ and $\hat{\Omega}^{HH}$ represent the inferred objective weights of the commercial bank and household agent.

Table 5.7 – Three-Agent IRL Model: Regime-Dependent Inferred Reward Weights $\hat{\Omega}$

	Full Sample (1990–2024)	Pre-2000 (1998–1999)	Lost Decade (2000–2012)	Abenomics (2013–2016)	YCC (2016–2022)
<i>BoJ – Policy Rate $\hat{\Omega}^{BoJ,rate}$</i>					
Inflation Gap	0.0008	−0.0131	0.0032	−0.0002	−0.0022
Output Gap	−0.0006	0.0592	−0.0030	−0.0024	0.0010
Capital Stress	−0.0109	0.0819	−0.0108	0.0054	−0.0243
Gov. Debt	−0.0008	−0.0255	0.0004	−0.0014	−0.0004
<i>BoJ – Quantitative Easing $\hat{\Omega}^{BoJ,QE}$</i>					
Inflation Gap	0.0013	0.0051	−0.0027	−0.0033	−0.0006
Output Gap	0.0008	−0.0555	0.0040	0.0052	−0.0030
Capital Stress	−0.0323	−0.0023	−0.0611	0.0352	−0.0036
Gov. Debt	−0.0051	−0.0381	−0.0025	−0.0026	0.0004
<i>Commercial Bank $\hat{\Omega}^{bank}$</i>					
NIM	−0.0222	−0.1178	0.0355	−0.0080	−0.0913
Capital Stress	−0.1248	−0.0283	0.1052	−0.0026	−0.1923
Default Proxy	−0.6705	−0.9024	−0.8193	−0.7906	−0.0080
Credit (HH)	0.4329	0.5676	0.0838	−0.0235	1.1778
<i>Household $\hat{\Omega}^{HH}$</i>					
Real Rate	−0.6837	−5.2450	−1.6369	9.3227	20.1063
Output Gap	−0.0943	0.1757	−0.1118	−0.0560	−0.0865
Inflation	−0.6703	−5.2532	−1.6157	9.2851	20.1595
Deposit Rate	0.0087	0.0255	0.0026	0.1398	−0.2173
HH Debt Proxy	0.0723	0.2727	0.0524	0.1598	0.0306

Notes: Regime-dependent inferred reward weights $\hat{\omega}$ for each agent, estimated separately over the full sample and four sub-periods: Pre-2000 (BoJ Zero Interest Rate Policy inception, 1998–1999); Lost Decade (2000–2012, encompassing QE1, Quantitative and Qualitative Easing transition); Abenomics (2013–2016, QQE with forward guidance); YCC (Yield Curve Control, 2016–2022). Intercepts excluded. Positive weights indicate reward-increasing state variables; negative weights indicate penalised states.

Table 5.8 – Three-Agent IRL Model: Full Parameter Dictionary (Global)

Symbol	Description	Value
<i>Output Gap Equation</i>		
α	Output gap persistence	0.7518
β	Effective credit	-0.0006
γ	Capital stress	-0.0003
<i>Inflation Equation</i>		
ρ	Inflation persistence	0.1633
λ	Output gap pass-through	26.9646
<i>Default Proxy Equation</i>		
δ_d	Default persistence	-0.2198
ϕ	Negative output gap	-12.3928
ψ	PNF credit	-0.0544
<i>Capital Stress Equation</i>		
κ	Capital stress persistence	0.0246
θ	Default proxy feed-through	1.2051
μ	NIM effect	-2.2835
<i>Household Debt Proxy Equation</i>		
δ_D	HH debt persistence	0.9892
ζ	HH credit effect	0.0232
ξ	Real rate effect	-0.0064
<i>Exogenous Constants</i>		
π^*	Inflation target (monthly)	0.0200
$\underline{\kappa}$	Capital stress floor	-2.0000
$\underline{D}$	HH debt ceiling	5.0000
<i>BoJ Reward Weights</i>		
ω_{π}^{rate}	Inflation gap (rate)	0.0008
ω_y^{rate}	Output gap (rate)	-0.0006
ω_{κ}^{rate}	Capital stress (rate)	-0.0109
ω_b^{rate}	Gov. debt (rate)	-0.0008
ω_{π}^{QE}	Inflation gap (QE)	0.0013
ω_y^{QE}	Output gap (QE)	0.0008
ω_{κ}^{QE}	Capital stress (QE)	-0.0323
ω_b^{QE}	Gov. debt (QE)	-0.0051
<i>Bank Reward Weights</i>		
ω_{κ}^{bk}	NIM	-0.0222
ω_{κ}^{bkm}	Capital stress	-0.1248
ω_{κ}^{bk}	Default proxy	-0.6705
ω_{κ}^{bk}	HH credit	0.4329
<i>Household Reward Weights</i>		
ω_r^{HH}	Real rate	-0.6837
ω_y^{HH}	Output gap	-0.0943
ω_{π}^{HH}	Inflation	-0.6703
ω_{dep}^{HH}	Deposit rate	0.0087
ω_D^{HH}	HH debt proxy	0.0723

Table 5.9 – Three-Agent IRL Model: Transition Dynamics Equations (YCC-Restricted)

Variable	Coef	Std Err	<i>t</i>	p-value	[0.025	0.975]
<i>Output Gap</i> _{<i>t</i>+1} $R^2 = 0.5914$						
Constant	−0.0007	0.001	−0.626	0.533	−0.003	0.002
Output Gap _{<i>t</i>}	0.7619	0.085	8.922	0.000***	0.591	0.932
Credit (PNF) _{<i>t</i>}	−0.0042	0.003	−1.634	0.107	−0.009	0.001
Capital Stress _{<i>t</i>}	−0.0009	0.000	−4.486	0.000***	−0.001	−0.000
<i>Inflation</i> _{<i>t</i>+1} $R^2 = 0.0039$						
Constant	0.3136	0.353	0.889	0.377	−0.390	1.018
Inflation _{<i>t</i>}	−0.0164	0.121	−0.135	0.893	−0.258	0.225
Output Gap _{<i>t</i>+1}	12.3981	24.451	0.507	0.614	−36.394	61.190
<i>Default Proxy</i> _{<i>t</i>+1} $R^2 = 0.0126$						
Constant	0.1537	0.045	3.431	0.001***	0.064	0.243
Default Proxy _{<i>t</i>}	−0.0774	0.133	−0.584	0.561	−0.342	0.187
Neg. Output Gap _{<i>t</i>+1}	−1.8714	3.649	−0.513	0.610	−9.155	5.413
Credit (PNF) _{<i>t</i>}	−0.0089	0.032	−0.283	0.778	−0.072	0.054
<i>Capital Stress</i> _{<i>t</i>+1} $R^2 = 0.0463$						
Constant	−0.7499	0.801	−0.936	0.353	−2.349	0.849
Capital Stress _{<i>t</i>}	−0.2089	0.121	−1.731	0.088*	−0.450	0.032
Default Proxy _{<i>t</i>+1}	0.7379	2.254	0.327	0.744	−3.762	5.237
NIM _{<i>t</i>}	−1.0552	8.537	−0.124	0.902	−18.095	15.984
<i>Household Debt Proxy</i> _{<i>t</i>+1} $R^2 = 0.9798$						
Constant	0.1136	0.045	2.524	0.014**	0.024	0.203
HH Debt Proxy _{<i>t</i>}	0.9737	0.017	55.661	0.000***	0.939	1.009
Credit (HH) _{<i>t</i>}	−0.0294	0.129	−0.228	0.820	−0.286	0.228
Real Rate _{<i>t</i>}	0.0097	0.014	0.671	0.504	−0.019	0.038

Notes: OLS estimates of the five transition equations re-estimated on the YCC sub-period (2016:02–2022:01). Structure identical to Table 5.5 (global model) to allow direct coefficient comparison. R^2 reported per equation. Standard errors are heteroskedasticity-consistent; 95% confidence intervals reported. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5.10 – Three-Agent IRL Model: Inferred Reward Functions $\hat{\Omega}$ (YCC-Restricted)

Variable	Coef	Std Err	t	p-value	[0.025	0.975]
<i>BoJ Policy Rate</i> $R^2 = 0.1279$						
Constant	0.0006	0.003	0.249	0.804	−0.004	0.006
Inflation Gap	−0.0019	0.003	−0.744	0.460	−0.007	0.003
Output Gap	0.0016	0.003	0.527	0.600	−0.004	0.007
Capital Stress	−0.0314	0.012	−2.656	0.010**	−0.055	−0.008
Gov. Debt	−0.0005	0.003	−0.204	0.839	−0.006	0.005
FX	−0.0004	0.003	−0.158	0.875	−0.006	0.005
VIX	0.0274	0.011	2.389	0.020**	0.004	0.050
<i>BoJ Quantitative Easing</i> $R^2 = 0.2939$						
Constant	0.0082	0.001	8.127	0.000***	0.006	0.010
Inflation Gap	−0.0007	0.001	−0.710	0.480	−0.003	0.001
Output Gap	−0.0041	0.001	−3.540	0.001***	−0.006	−0.002
Capital Stress	−0.0047	0.005	−0.993	0.325	−0.014	0.005
Gov. Debt	0.0004	0.001	0.433	0.666	−0.002	0.003
FX	−0.0014	0.001	−1.347	0.183	−0.003	0.001
VIX	0.0069	0.005	1.507	0.137	−0.002	0.016
<i>Commercial Bank</i> $R^2 = 0.8014$						
Constant	0.2980	0.081	3.658	0.001***	0.135	0.461
NIM	−0.0698	0.083	−0.844	0.402	−0.235	0.095
Capital Stress	−0.2483	0.087	−2.861	0.006***	−0.422	−0.075
Default Proxy	−0.0033	0.090	−0.037	0.971	−0.184	0.177
Call Rate	−0.0605	0.087	−0.698	0.488	−0.233	0.113
Credit (HH)	1.2385	0.099	12.513	0.000***	1.041	1.436
<i>Household</i> $R^2 = 0.2520$						
Constant	0.0587	0.035	1.671	0.100*	−0.011	0.129
Real Rate	20.6848	7.403	2.794	0.007***	5.905	35.465
Output Gap	−0.1175	0.035	−3.321	0.001***	−0.188	−0.047
Inflation	20.7135	7.401	2.799	0.007***	5.936	35.491
Deposit Rate	−0.0313	0.037	−0.838	0.405	−0.106	0.043
HH Debt Proxy	0.0319	0.037	0.863	0.391	−0.042	0.106

Notes: OLS estimates of the inferred reward functions $\hat{\Omega}$ re-estimated on the YCC sub-period (2016:02–2022:01). Structure identical to Table 5.6 (global model) to allow direct comparison. R^2 per equation. Standard errors are heteroskedasticity-consistent; 95% confidence intervals reported. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5.11 – Three-Agent IRL Model: Inferred Reward Weights $\hat{\Omega}$ (YCC-Restricted)

State Variable	$\hat{\omega}$
<i>Bank of Japan – Policy Rate</i> $\hat{\Omega}^{BoJ,rate}$	
Inflation Gap	−0.0019
Output Gap	0.0016
Capital Stress	−0.0314
Gov. Debt	−0.0005
<i>Bank of Japan – Quantitative Easing</i> $\hat{\Omega}^{BoJ,QE}$	
Inflation Gap	−0.0007
Output Gap	−0.0041
Capital Stress	−0.0047
Gov. Debt	0.0004
<i>Commercial Bank</i> $\hat{\Omega}^{bank}$	
NIM	−0.0698
Capital Stress	−0.2483
Default Proxy	−0.0033
Credit (HH)	1.2385
<i>Household</i> $\hat{\Omega}^{HH}$	
Real Rate	20.6848
Output Gap	−0.1175
Inflation	20.7135
Deposit Rate	−0.0313
HH Debt Proxy	0.0319

Notes: Inferred reward weights $\hat{\omega}$ re-estimated on the YCC sub-period (2016:02–2022:01). Intercepts excluded.

Table 5.12 – Three-Agent IRL Model: Full Parameter Dictionary (YCC-Restricted)

Symbol	Description	Value
<i>Output Gap Equation</i>		
α	Output gap persistence	0.7619
β	Effective credit	-0.0042
γ	Capital stress	-0.0009
<i>Inflation Equation</i>		
ρ	Inflation persistence	-0.0164
λ	Output gap pass-through	12.3981
<i>Default Proxy Equation</i>		
δ_d	Default persistence	-0.0774
ϕ	Negative output gap	-1.8714
ψ	PNF credit	-0.0089
<i>Capital Stress Equation</i>		
κ	Capital stress persistence	-0.2089
θ	Default proxy feed-through	0.7379
μ	NIM effect	-1.0552
<i>Household Debt Proxy Equation</i>		
δ_D	HH debt persistence	0.9737
ζ	HH credit effect	-0.0294
ξ	Real rate effect	0.0097
<i>Exogenous Constants</i>		
π^*	Inflation target (monthly)	0.0200
$\underline{\kappa}$	Capital stress floor	-2.0000
$\bar{D}$	HH debt ceiling	5.0000
<i>BoJ Reward Weights</i>		
ω_{π}^{rate}	Inflation gap (rate)	-0.0019
ω_y^{rate}	Output gap (rate)	0.0016
ω_{κ}^{rate}	Capital stress (rate)	-0.0314
ω_b^{rate}	Gov. debt (rate)	-0.0005
ω_{π}^{QE}	Inflation gap (QE)	-0.0007
ω_y^{QE}	Output gap (QE)	-0.0041
ω_{κ}^{QE}	Capital stress (QE)	-0.0047
ω_b^{QE}	Gov. debt (QE)	0.0004
<i>Bank Reward Weights</i>		
ω_{κ}^{bk}	NIM	-0.0698
ω_{κ}^{bkm}	Capital stress	-0.2483
ω_{δ}^{bk}	Default proxy	-0.0033
ω_{cHH}^{bk}	HH credit	1.2385
<i>Household Reward Weights</i>		
ω_r^{HH}	Real rate	20.6848
ω_y^{HH}	Output gap	-0.1175
ω_{π}^{HH}	Inflation	20.7135
ω_{dep}^{HH}	Deposit rate	-0.0313
ω_D^{HH}	HH debt proxy	0.0319